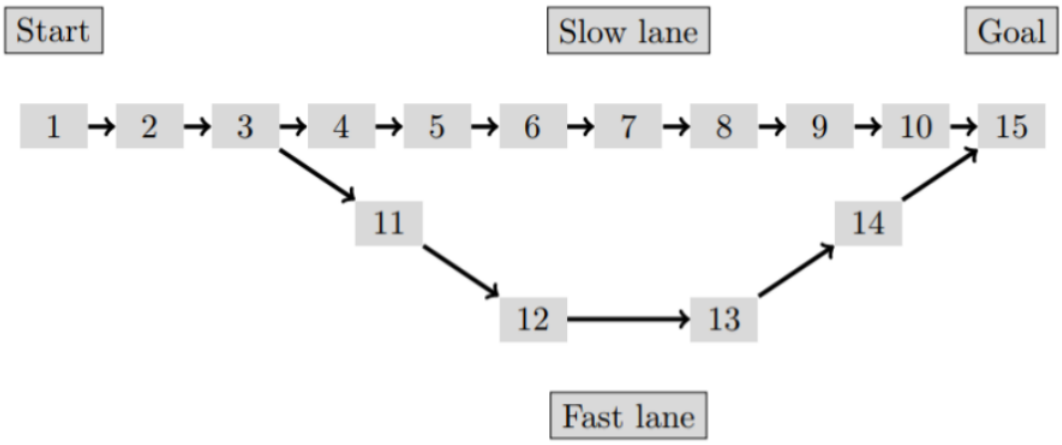
Academic year 2025-2026 Ecole Polytechnique de Louvain LINFO2275

LINFO2275
First project

Group 2:

Gavin Hope 09022500 – **SINF**
Jakob Thibodeau 12752500 – **DATI**
Lore Lernoux 77562000 – **MAP**



Professor: Marco Saerens

Introduction

In this project, we present a policy optimization process for a Markov Decision Process model of a simplified Snakes and Ladders game. To find this optimal policy, we implemented the value iteration algorithm [KWW22]. This policy specifies, for each tile, the best dice to use in order to minimize the expected number of turns to reach the goal (tile 15). In this game, the player may choose between four different dices at each turn, each with a unique set move and trap probabilities. There are two game configurations considered: one circular (must stop on tile `15` exactly) and one linear (must stop exactly on tile `15`). Afterwards, we compared this optimal policy to other suboptimal policies, such as fixed-dice policies, to compare performance in actual simulations of the game. Both theoretical results from value iteration and empirical results from Monte Carlo simulations are analyzed, presented and compared across different trap layouts, for both game configurations.

Theoretical background

The game is modeled as a Markov Decision Process. As seen during the lectures, it is defined by [Bra21]

- a finite set of states S, which in our case corresponds to the tiles of the board, i.e. $S = \{1, 2, \ldots, 15\}$;
- a set of actions A, which corresponds to the choice of dice to roll, i.e. $A = \{1, 2, 3, 4\}$;
- a transition function $P(s' \mid s, a)$, which gives the probability of reaching state s' from state s after choosing action a. These probabilities are determined by the stochastic outcomes of the dice, the branching structure of the board, and the activation of traps;
- a cost function $c(s \mid a)$, which gives the cost when taking action a in state s. In our case, we can define the cost as 1 for each turn taken, and 0 when reaching the goal (tile `15`).

The goal is to find an optimal policy π^* specifying, at each tile, which dice selection minimizes the expected number of turns to reach tile `15`. Let $V(s)$ be this total expected number of turns which respects Bellman's equation with as condition $V(15) = 0$:

$$V(s) = \min_{a \in A} \left[c(s \mid a) + \sum_{s' \in S} P(s' \mid s, a) V(s') \right]$$

This optimal expected number of turns can be computed using the value iteration algorithm, which iteratively updates $V(s)$ for each state until convergence. The optimal policy is then obtained by selecting, for each state, the action that minimizes the Bellman equation.

To validate the theoretical results, we performed Monte Carlo simulations. To do so, we simulated `1000` games to completion following each strategy and determined the average number of total turns, $\bar{T}$. We also computed the standard error of these estimates based on the sample standard deviation $\hat{\sigma}$.

$$\bar{T} = \frac{1}{N} \sum_{i=1}^{N} T_i$$

$$\text{SE} = \frac{\hat{\sigma}}{\sqrt{N}}$$

$\bar{T}$ is then compared to the theoretical value $V(1)$ obtained through value iteration in order to validate the correctness of the optimal policy.

Implementation description

Trap layouts

We chose to use 6 different trap layouts to compare the optimal policy in different settings. These layouts are the following:

- **No traps**: This layout has no traps, so it is the simplest layout and serves as a baseline for comparison.
- **Few traps**: This layout has several traps scattered throughout the board that move the player three squares backward.

- **Many traps**: This layout has a variety traps of all types scattered throughout the board, which makes it more difficult to win the game.
- **Evil fast lane**: This layout has restart and backward traps only on the fast lane.
- **Two-in-a-row traps**: This layout presents three segments with two consecutive backward trap tiles, making avoiding them more difficult.
- **Back-to-three traps**: This layout has traps that send the player back to tile 3, which is the branching tile.

Sub-optimal strategies

We chose to implement five sub-optimal strategies to compare with the optimal policy. These strategies are the following :

- **One dice only**: For each of the 4 dices, this policy simply uses one dice type. This gives us 4 sub-optimal policies to compare with the optimal one.
- **Uniform random strategy**: This policy chooses a dice uniformly at random, with no regard for the current tile. This is a baseline strategy.

Results

Optimal Policy

For each trap layout defined above, we graphically represented the board and its optimal policy by coloring the tiles according to the policy dice decision. The shape of the tile indicates which type of trap is present on it (if any), and the number in each tile indicates the approximate theoretical expected number of turns to win the game from that tile when following the optimal policy. For each trap layout, we represented both the circular and linear game structures, to see how the optimal policy changes with both a circular and non-circular layout.

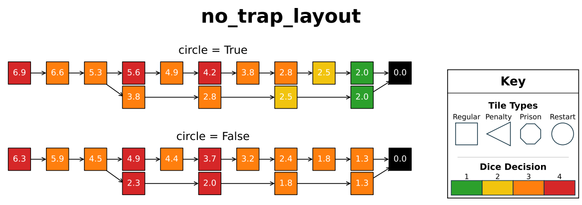


Figure 1: Expected number of turns to win the game from each tile for the no traps layout.

Figure 1 shows the optimal policy in the case where no traps are present, in both the circular and linear game structures. For tiles closer to the goal, the circlular structure favors safer dice as it must stop exactly on the goal, whereas the linear structure favors faster dices as it can still win by overshooting the goal. For the rest of the tiles, the policy mostly favors dice `3`, as it has the highest expected movement value (`1.5`) of all dices, assuming traps are not triggered. Dice `4` (the special dice) is chosen on tile `1`, since the player cannot move further back, bringing the expected value of the dice to `2`. Tiles `4` and `6` also favor the special dice, since there is a $\frac{1}{4}$ chance to move back directly to tile `3` and potentially move to the fast lane. It is surprising that tile `3` itself does not favor the special dice, as it has the possibility of landing directly land on the goal tile. This is likely due the low probabilty (12.5% chance) of this outcome as it both taking the fast lane (50% chance) and also rolling a 5 (25% chance).

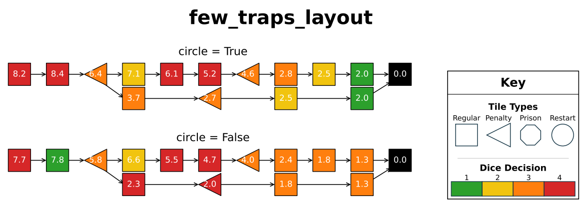


Figure 2: Expected number of turns to win the game from each tile for the few traps layout.

Figure 2 adds a few penalty traps into the layout. We see some recurring patterns such as safer dices being advantageous closer to the goal for the looped layout and prioritizing the riskier dices in safer regions. On most trap tiles, the optimal policy chooses the risky dice even with the $\frac{1}{4}$ chance of activating the trap. This can be explained by comparing the expected displacement: even with the penalty traps effectively acting as -3 rolls when landing on them, the risky dice has an expected value of $\frac{3}{4}$, which is higher than the safe and normal dices ($\frac{1}{2}$). More interestingly, the optimal policy clearly tries to avoid the traps when it can by anticipating them. For example, it chooses the normal dice on tile 4 to get closer to the trapped tile 7 until it is close enough (on tile 5 or 6) to be able to leap over the trap using the special dice.

The non-looped layout prioritizes heavily taking the fast lane, as seen by the choice of the safe dice on tile 2. Even if taking the fast lane is not guaranteed, if there is even a slight chance of going there, it takes it. In practice, the player will only rarely take the fast lane with this policy since the only way to end on tile 2 is to roll a -3 when standing on tile 5. Still, when it happens, it is a huge speedup.

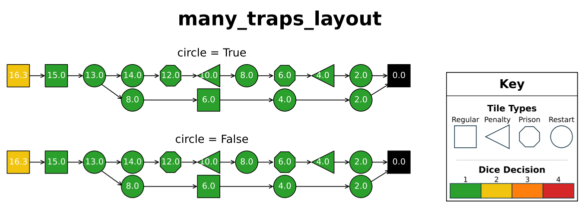


Figure 3: Expected number of turns to win the game from each tile for the many-trap layout.

As see in Figure 3, having many traps unsurprisingly leads to the near exclusive use of the safe dice. As the safe dice moves the player by at most 1 tile, having the layout loop or not doesn't affect the optimal policy choice or its effectiveness. This is further shown by comparing to the only dice 1 strategy, which takes almost the same number of turns as the optimal strategy.

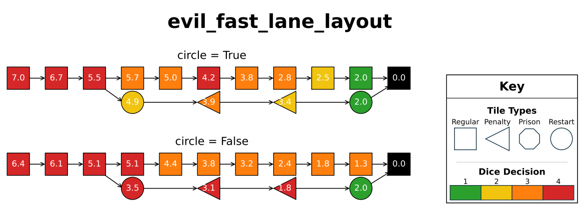


Figure 4: Expected number of turns to win the game from each tile for the evil fast lane layout.

When designing the *evil fast lane* layout seen in Figure 4, we expected the optimal policy to avoid the fast lane at all costs. However, that is not what we observe, as the choice of the special dice at tile 7 for the looped layout and tile 4 for the non-looped layout may directly lead the player to land on tile 3 and merge into the fast lane. This is an acceptable outcome for the model, since rolling a 5 with the special dice from tile 3 would lead directly to the goal tile when taking the fast lane.

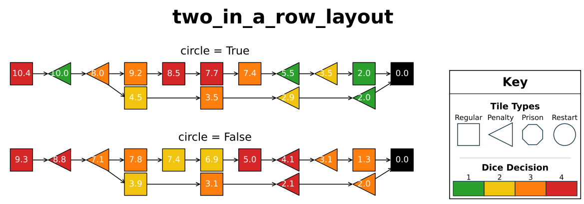


Figure 5: Expected number of turns to win the game from each tile for the two-in-a-row traps layout.

In the layout shown in Figure 5, we observe a drastic difference in strategy between the looped and unlooped boards. In the looped board, the player tries to leap the traps, but only from tiles that guarantee it can land on or before the goal if it rolls a 5, as shown by the choice of the special dice on tiles 5 and 6. However, in the unlooped layout, the player prefers inching closer using the normal dice, even risking the activation of the trap on tiles 8, 13 and 14, so that a 3 or a 5 on the special dice makes it reach the goal (skipping over the 2 traps in a row).

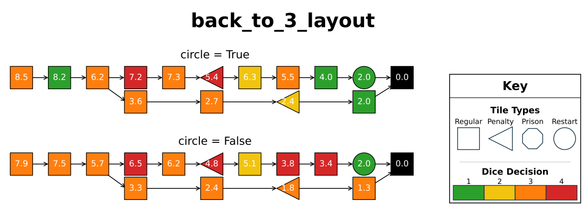


Figure 6: Expected number of turns to win the game from each tile for the back-to-three traps layout.

In the back-to-3 layout shown in Figure 6, the optimal strategy becomes much more complex. In the looped layout, the policy would much rather take the fast lane (as evidenced by the choice of the safe dice on tile 2) since it avoids having to skip past the restart trap on tile 10. Surprisingly, the optimal strategy chooses the risky dice at tile 8, risking triggering the restart trap. It appears that the risk of restarting is worth taking over moving slowly (with the safe dice) to the goal. The player in the non-looped layout does not benefit as much from the fast lane, as it can simply leap over tile 10 to reach the goal with the special dice.

Theoretical vs Empirical Results: Sub-optimal Strategies

We compared the optimal policy with the five sub-optimal strategies by simulating 1000 games for each policy and game structure (circle or not) with Monte Carlo simulations. We also included the theoretical expected number of turns, $V(1)$, for each strategy computed by value iteration. In all experiments, empirical means are very close to the theoretical values, indicating the convergence of the value iteration algorithm. Note that, for some layouts that are not shown in this report, no convergence could be achieved for any strategy (including the optimal one), as the player would get stuck in traps and never reach the goal. This, for example, could happen on a Dice 4 only policy where restart and penalty traps are present on the five first tiles, as the player will always get teleported back to the start of the board and never reach the goal, leading to an infinite expected number of turns.

Figure 7 compares different strategies to the optimal policy in the no-traps layout.

In the circle layout, the optimal policy far outperforms the other simpler strategies, since using riskier dice first and safer dice near the goal brings a huge advantage to the player. For the non-circle layout, the advantage is less significant, since repeatedly rolling dice 3 is nearly the optimal policy anyway. Note that for all layouts, looped or not, the *safe dice only* strategy has an expected number of turns of 17. This value remains the same, as the safe dice never triggers any traps and can move the player at most 1 tile, never overstepping the goal.

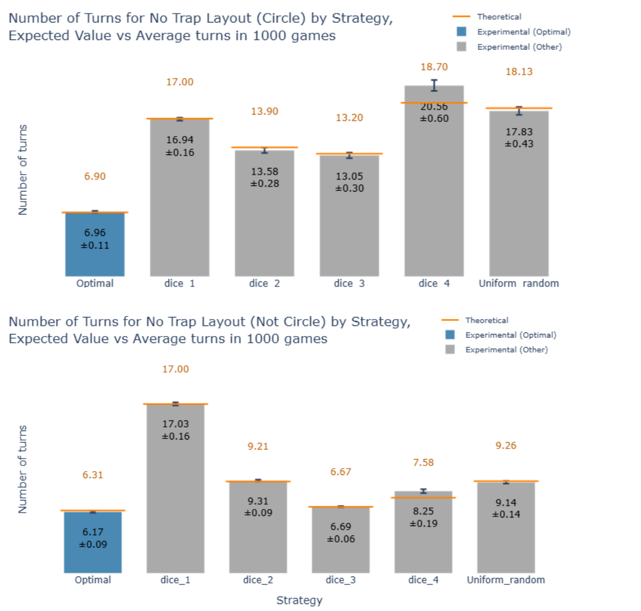


Figure 7: Expected vs Average simulated number of turns to win by strategy for No Traps Layout

Comparing the optimal policy to other strategies for the few-trap layout, as seen in Figure 8, we observe mostly the same trends as with no traps. However, since the use of riskier dice is now punished, the strategies solely using them are worse, digging the optimal policy's lead even further. Indeed, a more heterogeneous use of the different dice is required to get a good performance, as the player needs to be able to adapt to the traps and use the safe dice when needed, while still taking risks when it is worth it. The *uniform random* strategy has worse performance than the optimal policy, but close to all other strategies,

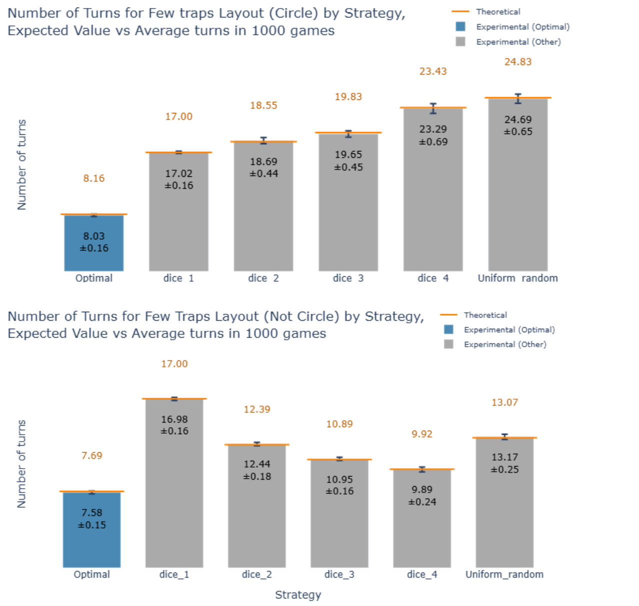


Figure 8: Expected vs Average simulated number of turns to win by strategy for Few Traps Layout

which was expected since it has a $\frac{1}{4}$ chance of using each dice at each tile.

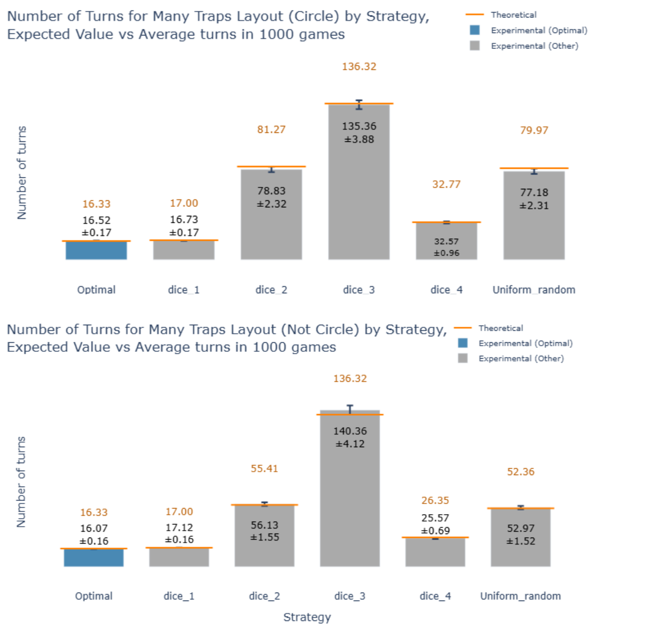


Figure 9: Expected vs Average simulated number of turns to win by strategy for Many Traps Layout

Looking at Figure 9, the optimal policy's performance is comparable to the *safe dice only* strategy since the optimal policy mainly uses the safe dice only. All other strategies, especially the *risky dice only* strategy, are penalized heavily by the traps. The *special dice only* strategy performs a little bit better, since it can leap over multiple traps in a single roll, and the traps do not activate at once in the same pass. Variance also increases for risky strategies compared to previous layouts, as the player is more likely to encounter traps and face unpredictable outcomes. Choosing a dice uniformly at random performs better than the risky dice only strategy, since it has a $\frac{1}{4}$ chance of using the safe dice, but it is still much worse than the optimal policy, which uses it almost exclusively.

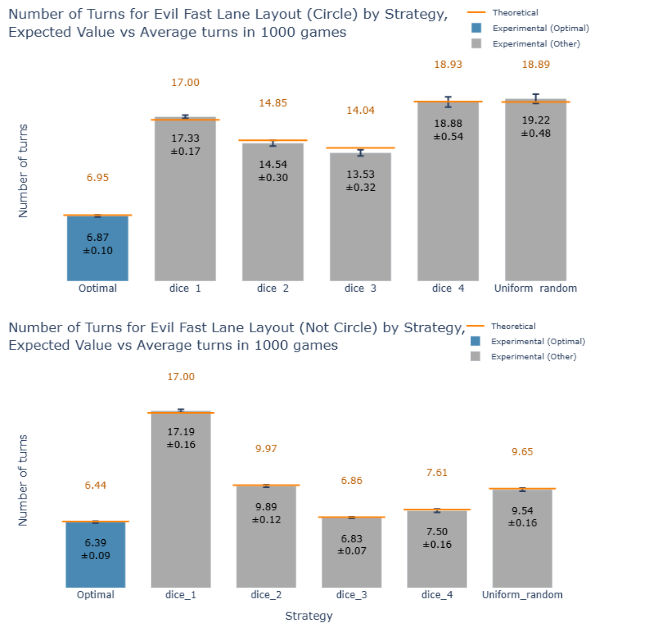


Figure 10: Expected vs Average simulated number of turns to win by strategy for evil fast lane Layout

The comparisons in Figure 10 continue to show the dominance of the optimal strategy, with the number of turns taken by all strategies closely following that of the no-trap layout.

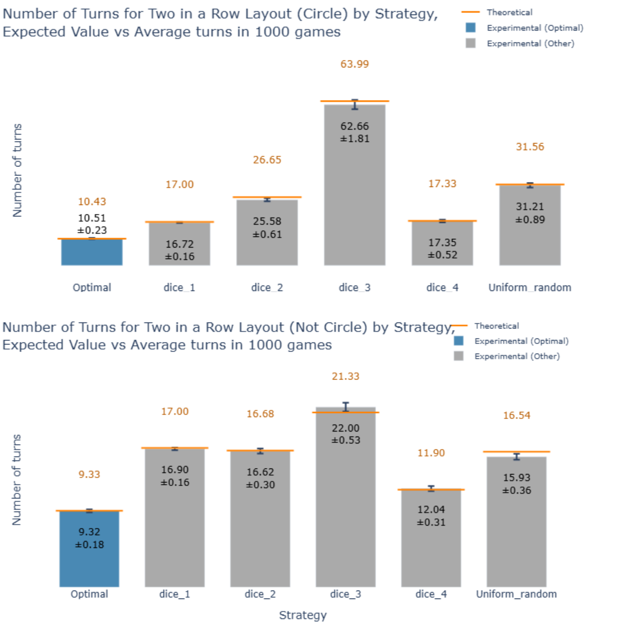


Figure 11: Expected vs Average simulated number of turns to win by strategy for two-in-a-row Layout

Looking at the comparisons in Figure 11, we find that the risky dice is heavily penalized if misused, since the chance of trap activation is much higher. Surprisingly, for the looped version, the *special dice only* strategy performs better on this layout than on the no-trap layout. This is due to the trap on tile 9, which if activated, leads the player back to tile 6, from which a roll of 5 directly lands on the goal. Compare that to the no-trap layout, where landing on tile 9 would mean a roll of 3 or 5 overshoots the goal and a roll of -1 or -3 moves the player back (like a trap) but consumes an extra turn (unlike a trap).

Figure 12 shows that having the layout loop or not clearly affects how risk-averse the player can be, as the *risky dice only* and the *special dice only* policies are nearly twice and thrice as fast, respectively.

Overall, value-iteration consistently provided the least number of turns compared to other more naive strategies. It accomplished this with, at least for this relatively small problem, usually below 50 iterations. In problems with simple models such as this one, a tighter stability tolerance is reasonable given the fast per-iteration calculation time. For more complex problems with a high cost per iteration, either computationally or monetarily, a larger stability tolerance could be chosen to determine a less optimal policy faster [**granzotto2020stopvalueiterationstability**].

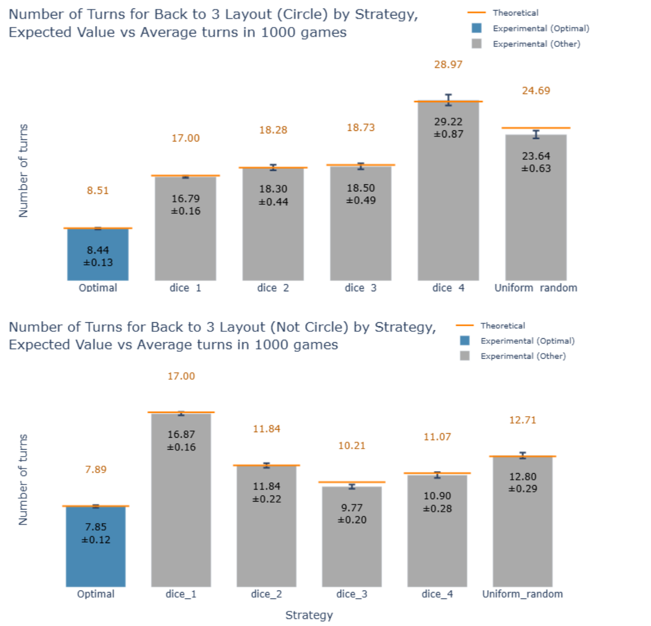


Figure 12: Expected vs Average simulated number of turns to win by strategy for back-to-three Layout

By selecting a larger stability tolerance, the number of iterations can be decreased in exchange of a suboptimal but quicker to determine policy. This makes value-iteration a flexible tool for dealing with MDPs of all problem sizes. Another factor to consider is problem complexity. For simple cases such as *many-traps* the naive solution performs comparably to the optimal strategy, with no value-iteration step required.

Conclusion

Overall, the optimal policy obtained through value iteration consistently outperformed the suboptimal strategies across all trap layouts and game configurations. The advantage of the optimal policy was particularly significant because of the heterogeneous way to use dices, as using riskier dices in the beginning and safer ones later. The Monte Carlo simulations validated the theoretical results, as the empirical average number of turns closely matched the expected number of turns computed by value iteration for all strategies and layouts. This confirms the correctness of our implementation and the effectiveness of the optimal policy in minimizing the expected number of turns to win the game.

References

[Bra21] Paolo Brandimarte. "From shortest paths to reinforcement learning". In: Springer, 2021. Chap. 3.1, pp. 68–74.

[KWW22] Mykel J. Kochenderfer, Tim A. Wheeler, and Kyle H. Wray. "Algorithms for Decision Making". In: MIT Press, 2022. Chap. 7, pp. 141–144.